

Computer Graphics and Human Simulation

This paper focuses on how to simulate human's features and behaviors through algorithms and computer graphics techniques that throughout the years have been improving and developing a new way to see and modify our surrounding in a virtual environment, better known as virtual world. The base of this research has been obtained from different sources that will help users understand the importance of human simulation in our world, as well as how the influence of such could help us to improve our world by making models and modification that would improve our environment. The topics of these resources can be better explained as Motion and Modeling Systems, Expressive Animation, Texture, Features, and Editing Systems.

Motion and Modeling Systems: this is a very important step to make computers simulate human features and behaviors and, the study of this can be broken into the following parts for a better understanding. Human Motion, according to my research to obtain a well defined given sequence of 3D motion capture data of human body, the 3D markers are a good tool. Markers are organized into hierarchy where each node corresponds to a meaningful part of human body. Motion Transition, on this report a series of experiments were studied to understand and investigate how some features can influence human transition. A good analysis found is a crowd, where motion algorithms are used. This analysis can help to improve the perceptual realism by effectively disguising the motion transitions.

Expressive Animation: Based on my research, one way to approach expressive animation for human simulation is by using a expressive speech and its emotion modifiers as input. The algorithm study on this report is a constrained dynamic implementation that searches for the best-matched captured motion clips, minimizing cost function, from a database.

Features: Head Motion, the studied applied on this topic was based on a presented novel data-driven approach to synthesize appropriate head motion by sampling from trained Hidden Markov Models. This technique has been used in first ordered models, where some techniques are used to smooth the synthesized sequence. Facial Motion, my study consists only on a data-driven 3D facial motion capture to system that use an orthogonal-blend-shape face model and constrained weight propagation.

Texture Synthesis: for eye motion, a technique found during my research was a data-driven texture approach of synthesizing realistic eye motion. The assumption is that the eye gaze may have a connection with eyelid motion as well as with the head motion and speech. Another approach statically modeling the innate coupling between gazes and head movements, this approach is said, according to my research, to generate natural gazes correlated with head motions.

Editing Systems: for speech animation, a different approach was studied and this is based on a co-articulation model with compact size that contains facial motion capture data. This model is applied to resulting phoneme sequence, producing a natural and detailed motion.

Finally, my research concluded in the study of how to create personalized facial avatars by using portraits and photos examples in an emulating real-world face sculpting process. This representation was based on a system to intuitively craft personalized 3D avatars by using 3D portraits or photos and sculpting the object shape towards any individual. This is achieved by dynamically using crafting algorithms to automatically generate crafted models using image examples-based painting algorithm. As a result, this final approach, along with the other techniques found during my research, can lead you to a good understanding and idea of how to simulate human behaviors and features through computer graphics techniques. Therefore, the implementation of each one of these could be combined to created a well function system that simulates humans.